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Machine Learning Assisted Selective Configuration Interaction for Accurate Ground and Excited State Calculations

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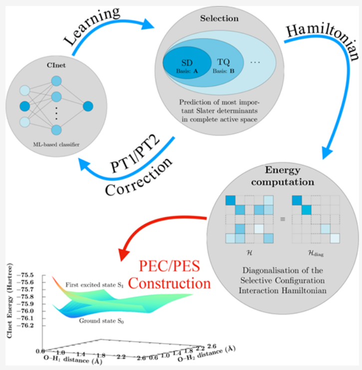
ABSTRACT: In this work, we introduce a perturbative Selective Configuration Interaction (SCI) approach guided by a binary machine-learning classifier. The method leverages a lightweight feedforward neural network (FNN), purposefully designed for fast and efficient training. Within this framework, we show that our model attains the accuracy of the state-of-the-art SCI method CIPSI (Configuration Interaction using a Perturbative Selection done Iteratively). Across all tested configuration-space sizes, our approach reliably identifies the most important Slater determinants using a binary cross-entropy metric, achieving FCI/CASCI-level accuracy within 10^{-4} Hartree for both ground and excited states. Finally, the method remains robust for strained geometries and conformational changes, successfully classifying Slater determinants across multiple points of potential energy curves and surfaces. This capability opens the door to new regression-based strategies for molecular electronic structure and the construction of machine-learning-driven potential energy surfaces.



1. INTRODUCTION

In many-electronic systems, electronic correlation plays a central role in determining total energy. This correlation is essential for obtaining a proper description of the true electronic density distribution, which governs all physical and chemical properties. The most straightforward approach to capturing this correlation is the Full Configuration Interaction (FCI) method.[1]

Conceptually, this method is simple. It involves expressing the total electronic wave function using the basis of all allowed Slater determinant:

$$|\Psi_{el}\rangle = c_0|\Psi_0\rangle + \sum_{a,r} c_a^r|\Psi_a^r\rangle + \sum_{ab,rs} c_{ab}^{rs}|\Psi_{ab}^{rs}\rangle + \sum_{abc,rst} c_{abc}^{rst}|\Psi_{abc}^{rst}\rangle + \cdots \quad (1)$$

where $|\Psi_0\rangle$ represents the Hartree−Fock determinant, and $|\Psi_a^r\rangle$, $|\Psi_{ab}^{rs}\rangle$, $|\Psi_{abc}^{rst}\rangle$, ⋯ denote the singly, doubly, triply (and so on) excited determinants constructed from the Hartree−Fock molecular orbitals (MO).

The objective is then to determine the coefficients c associated with each determinant, which can be achieved by diagonalization of the FCI Hamiltonian. However, due to the combinatorial growth of the FCI basis with respect to the number of MO, the FCI matrix rapidly becomes extremely large, which hinders the practical usefulness of this method. The FCI wave function in eq 1 is thus often truncated and limited to low-rank excitations. For instance, the CISD method includes only singly and doubly excited determinants. However, this truncation, which approximate the electronic correlation, might lead to poor description, particularly at the dissociation limits where the degeneracy of electronic configurations leads to strong correlation.[2]

Other approaches involve selecting the most relevant determinants, a strategy known as Selective CI (SCI) methods. Unlike the CISD method, where Slater determinants are chosen a priori based on excitation criteria, these methods select configurations over the entire set based on their estimated contribution to the FCI wave function.

One notable example is the Configuration Interaction using a Perturbative Selection done Iteratively (CIPSI) algorithm initially introduced by Huron *et al.*[3] Starting from a given variational Slater determinant subspace $|\Psi^{(0)}\rangle = \sum_i c_i|\phi_i\rangle$, the main idea is to iteratively select external determinants $|\phi_j\rangle$ using a second-order perturbative criterion (PT2). If the corrected energy contributions of the selected determinants exceed a given threshold, they are added to the initial subspace, and the procedure is repeated until the desired convergence is reached. Since its introduction in 1973, the CIPSI method has undergone numerous improvements and is now implemented as a semistochastic algorithm (ensuring more scalability) in the Quantum Package program.[4] This selective approach inspired us to develop our model.

Since the pioneering SCI approach, many other techniques have been developed. For instance, Adaptative CI[5,6](ACI)

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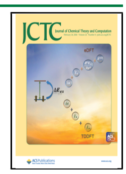

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selects new configurations based on energetic and perturbative criteria, but only those directly connected to the most important configurations in variational space. Other approach enriches the wave function using heat-bath sampling, where coefficients are evaluated through a first-order perturbation (PT1).[7]

For practical reasons, applying CI or SCI methods often requires reducing the number of configurations. To this end, an active space composed of a selected subset of occupied and virtual molecular orbitals is defined. The Complete Active Space (CAS) approach constructs the wave function as an expansion over all Slater determinants built from these chosen orbitals. Similar to FCI, the CASCI method optimizes all CI coefficients involved in the complete expansion of the wave function (eq 1). We can also mention the CAS Self-Consistent Field (SCF) method, in which both the CI coefficients and the molecular orbitals are optimized simultaneously.[8]

These methods are highly valued—as alternatives to traditional CI approaches such as CISD—when dealing with large polyelectronic systems or strongly correlated wave functions, for example in bond-breaking processes. In the literature, we commonly encounter active spaces of widely varying sizes, involving from few hundred determinants to several million, depending on the complexity of the problem.[9,10]

To our knowledge, the largest CI expansion in a CASCI calculation contained 10^{12} determinants, representing 22 electrons distributed across 22 molecular orbitals (22e, 22o).[11] This limit was achieved using highly efficient CI algorithms with massive parallelization. More recently, Gao *et al.*[12] surpassed the one-trillion-determinant record using a distributed implementation.

The development of Machine Learning (ML) techniques has the potential to redefine how such problems are addressed. In recent years, numerous ML models have been proposed and successfully applied to molecular dynamics and molecular property prediction, with the goal of accelerating quantum chemical computations.[13−17] These approaches rely on the statistical distribution of data generated at a chosen quantum chemical level of theory, such as Density Functional Theory (DFT).[18] From an initial data set, a model—such as a kernel or a neural network—is trained and subsequently applied to predict properties of a second data set.[19] The efficiency of these techniques is underpinned by the universal approximation theorem,[20] which establishes that neural architectures can approximate arbitrarily complex, high-dimensional functions.

Building on this assumption, Carleo and Troyer[21] proposed a Restricted Boltzmann Machine (RBM) model to learn the coefficients involved in quantum states in an unsupervised manner. In this novel approach, the RBM model serves as a wave function ansatz, and the learning process is optimized using a variational Monte Carlo method. Compared to traditional methods, the authors demonstrated that for spin models, such as Ising and Heisenberg Hamiltonians, the RBM model yielded lower variational energies with a reduced computational complexity.

Choo *et al.*[22] extended this preliminary work to realistic electronic Hamiltonians. Inspired by quantum information encoding, they mapped Fermionic models onto spin-like Hamiltonians using the Jordan-Wigner transformation.[23] This transformation demonstrated a strict equivalence with the FCI method. As previously discussed, FCI coefficients were learned via Monte Carlo sampling and the variational principle.[24] While chemical accuracy in energy was achieved for small basis sets, the convergence of variational Monte Carlo (VMC) could be slow for larger basis sets due to the influence of the most drawn Hartree−Fock state. A more promising method appears to be the one proposed by Herzog *et al.*,[25] which developed a RBM model in a generative context.

The inherent limitation of the VMC method can be overcome by employing machine learning regressive approaches. However, since electronic many-body wave functions are not smooth in configuration space, these regressions could be challenging. Despite this, the flexibility of deep neural networks is now being leveraged to represent electronic structures in real space, as demonstrated in refs26−31. However, these ML architectures are often complex and involve optimizing a large number of parameters.

Since 2018, Coe[32,33] introduced a semirandom selective method driven by neural networks. In his paper, he employed a regression approach to learn a training set of determinants and coefficients through minimization of the Root Mean Square Error (RMSE). This training model was then used to predict the strength of randomly selected determinants, which were added to the current variational space in a similar manner to the CIPSI procedure. This method was shown to accelerate the convergence of the traditional Monte Carlo CI. On the other hand, to classify wave function, Pineda Flores[34] chose to use a Support Vector Machine (SVM) with a Gaussian kernel (ChemBot). More recently,[35] a Convolutional Neural Network (CNN) model incorporating pseudopotentials with frozen core electrons was tested and demonstrated strong performance in predicting FCI energies along binding curves. While these approaches achieve good convergence in terms of total energy, they have not been proven to be efficient in terms of excited states.

Jeong *et al.*[36] compared various machine learning classification approaches in an Active Learning context (ALCI). They applied these models to first neutral excited energy of Polycyclic Aromatic Hydrocarbons (PAH) and demonstrated their ability to significantly reduce the active space size. For hexacene described through an active space composed of 26 electrons and 26 molecular orbitals (26e, 26o), they reduced the initial full space of 10^{14} determinants to only 10^4. However, we observed a significant deviation from CASCI reference calculations. For the three compared systems (naphthalene, anthracene, and pyrene), the error was approximately 0.3 eV ($\sim 10^{-2}$ Hartree, ~6.0 kcal/mol).

In this work, we focus on employing a lightweight Machine Learning architecture for Selective CI applications. This choice is motivated by the need for an easy-to-train model, as training can be time-consuming when dealing with large data sets. Additionally, since the proposed algorithm is based on Active Learning, the training function is often called, making a small model size advantageous.

The model named CInet was tested on a large set of chemical compounds (CH_4, C_2H_2, C_2H_4, C_2H_6, H_2O, NH_3, HCN, HF, benzene and naphthalene) described in different basis set functions (STO-3G, 6−31G, and cc-pVDZ) and compared with standard calculations: CIPSI and FCI/CASCI. Our energies rapidly achieve the FCI/CASCI energy with an accuracy below 10^{-4} Hartree (~0.06 kcal/mol, 0.003 eV). We also demonstrated the model's robustness by applying it to first excited states and constrained geometries, including potential energy curves and surfaces (PEC and PES). To our knowledge,

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no existing ML-based selective method has been shown to reach such accuracy for both ground and excited states. In particular, the efficiency of our approach stems from its ability to train all states simultaneously while using independent neural networks for each of them.

The paper is organized into two sections. The first section delves into the computational details, including the ML descriptors used to represent Slater determinants, the ML model setup, and the selection algorithm employed to identify the most significant electronic configurations. The second section presents the results, comparing the ML technique with reference methods CIPSI and FCI/CASCI through the computation of excited states, and PEC and PES.

2. COMPUTATIONAL METHOD

CInet is a homemade Python code using the PySCF[37,38] and SciKit-Learn libraries.[39] It performs all electronic integrals computations and diagonalization processes through modified PySCF subroutines. The code is available on GitHub (b4st13nc/cinet).

2.1. Representation of Slater Determinants

In common ML approaches, each data point is represented by an input vector, where each component defines a feature space. The choice of algorithm is important, but the representation of the data is equally crucial for achieving desired performance. For example, the review by Musil *et al.*[40] demonstrates the variety of descriptors used to encode physical and chemical information. The representation of knowledge can significantly differ based on the targeted properties, as illustrated in refs41,42.

This study follows the approach proposed by Coe in his MLCI method.[32] The feature space representing Slater determinants is based on the occupation number of spin orbitals. Each component is assigned a value of one if the corresponding spin orbital is occupied and zero otherwise. This allows each determinant to be represented as a binary input vector. Each binary vector is constructed by concatenating the α and β spin configurations, which represent all possible ways to combine N spin electrons among K spin orbitals. By convention, the α part represents all the spin-up electrons, while the β part represents all the spin-down electrons. For instance, in the case of 14 electrons distributed in 10 molecular orbitals (14e, 10o)—*i.e.*, 20 spin orbitals—, the Hartree–Fock determinant is then represented for the neural network as follows:

$$|\psi_0\rangle \equiv \underbrace{(1, 1, 1, 1, 1, 1, 1, 0, 0, 0}_{\text{spin-orbitals } \alpha} \underbrace{, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0)}_{\text{spin-orbitals } \beta} \quad (2)$$

where all the lowest α and β spin–orbitals are occupied.

This study exclusively focuses on spin-restricted systems, meaning the number of electrons and spin orbitals is identical for the α and β components. These configurations are called *strings* and are represented by a 64-bit integer. To minimize memory usage, the determinants are generated on-the-fly based on the evaluated excitations in the active space.

After establishing descriptors for the Slater determinant objects, we now turn to the second essential component of a ML approach: the model.

2.2. The Neural Network Model

The probability distribution of determinants $|\phi_i\rangle$ is not smooth in configuration space. In fact, two nearby determinants in terms of occupation—*e.g.*, $|\phi_1^4\rangle$ and $|\phi_1^5\rangle$, where a hole is created in the first spin orbital and a particule is added in the fourth or the fifth spin orbital—may not contribute the same in the total wave function. Consequently, a regressive-based approach may not be well-suited for this space.

Inspired by Jeong *et al.*,[36] we adopted a classification method. In our approach, Slater determinants are divided into two classes: important and unimportant. This dichotomy is based on the absolute value of coefficients $|c_i|$ and a specific threshold $|c_{\text{lim}}|$, above which determinants are considered important:

$$\tilde{c}_i = \begin{cases} 1 & \text{if } |c_i| > |c_{\text{lim}}| \quad (\text{important}) \\ 0 & \text{if } |c_i| < |c_{\text{lim}}| \quad (\text{unimportant}) \end{cases} \quad (3)$$

This parameter is essential for tuning the desired accuracy. The influence of this value on the model's performance is extensively discussed in the Supporting Information. For accurate calculations, we recommend setting the default value of $|c_{\text{lim}}| = \frac{1}{N_{\text{det}}}$, where N_{det} represents the total number of determinants in the active space.

For these two classes, the model's learning process is based on minimizing the binary cross-entropy cost function, which is commonly used in classification problems. This function is defined as

$$\mathcal{L}(\mathbf{y}, \hat{\mathbf{y}}) = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} y_i \cdot \log \hat{y}_i + (1 - y_i) \cdot \log(1 - \hat{y}_i) \quad (4)$$

where $\mathbf{y}$ represents the vector containing the N_{train} reference binary values of the training set—*i.e.*, the transformed Slater determinant coefficients $\tilde{c}_i$—and $\hat{\mathbf{y}}$ the corresponding binary values predicted by the neural network.

It has been shown that Feedforward Neural Network (FNN) architectures are more efficient than kernel-based methods or decision trees.[36] For this assessment, we used a simple FNN with a single hidden layer, as illustrated in Figure 1. To determine the most effective FNN model, we tested three crucial hyper-parameters: the number of neurons, the activation function applied to them, and the regularisation factor. We evaluated the model's performance on a single system (H_2O/6–31G) with over a million determinants.

Due to the difficulty in measuring the efficiency of a classification model, we decided to use an average of three metrics: the accuracy score σ, the phi coefficient ϕ, and the recall ρ. These metrics are summarized and detailed in the Supporting Information. After all realized calculations, we concluded that a FNN composed of 250 neurons activated by a ReLU function was the most effective model. During training, we found that the optimal regularisation factor was 10^{-2}.

In the next section, we will explain how the model is trained and used to select the most important configurations involved in the FCI/CASCI wave function.

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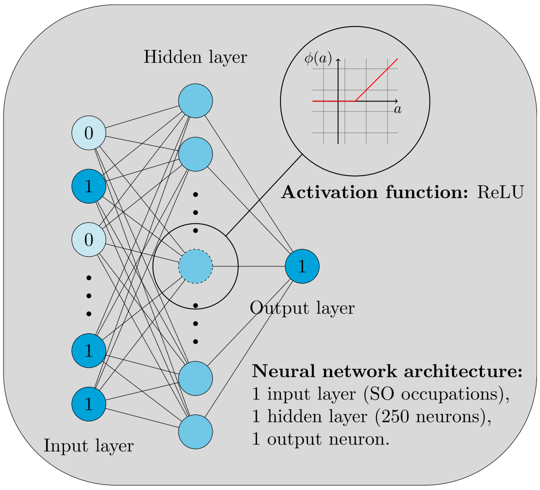

Figure 1. Illustration of the Feedforward Neural Network (FNN) used in the CInet algorithm.

2.3. The CInet Algorithm

In the CInet algorithm (Figure 2), the training set is initially defined through a CISD calculation. As presented in previous sections, the coefficients of the ground state CISD eigenvector (or any other excited state eigenvector) are transformed to organize the contributing determinants into two classes. Subsequently, the FNN is trained on this restricted configuration space (Space: **A**) by minimizing the binary cross-entropy defined in eq 4.

Once the neural network is trained, predictions are made on the adjacent subspace of configurations. This subspace is constructed by combining the single and double excitations of the current subspace. For example, if we start within the single and double configuration space, the adjacent space includes all triple and quadruple excited determinants. Due to the potential size of the new space, the total number of determinants is split into randomized batches of 100,000 configurations to reduce memory usage. At this stage, the selected determinants form a new basis (Space: **B**).

Similar to the CISPI approach, the energy contribution of new determinants is estimated using a second-order perturbative (PT2) calculation:

$$E_{\mathrm{PT2}} = \sum_{j \in \mathbf{B}} \frac{\left(\sum_{i \in \mathbf{A}} c_i H_{ij}\right)^2}{E_0 - H_{jj}} \tag{5}$$

where E_0 and c_i represent the energy and coefficients of determinants in variational space **A**, respectively. H_{jj} and H_{ij} are the Hamiltonian matrix elements computed using PySCF electronic integrals.

Since all ML models present an intrinsic error, predictions are always incorrect. This means that the space **B** can contain false positives (incorrectly identifying important determinants) or miss important ones. To exploit this, an active learning loop of ten epochs was incorporated. To guide this learning process, the coefficients c_j of new determinants are approximated using a first-order perturbation:

$$c_j = \frac{\sum_{i \in \mathbf{A}} c_i H_{ij}}{E_0 - H_{jj}} \tag{6}$$

When the relearning is achieved, the algorithm reiterates this procedure over each batch until PT2 energy becomes less than a particular limit fixed as

$$\epsilon_{\mathrm{lim}} = \frac{1}{\widetilde{N}_{\mathrm{det}}} \tag{7}$$

where $\widetilde{N}_{\mathrm{det}}$ represents the total number of determinants in the current adjacent configuration space. In fact, the more the space is large, the more the small accumulated contributions can affect significantly the total energy. To summarize, the cinet algorithm constructs the electronic wave function using an iterative cycle combining prediction and relearning.

At the end, all selected determinants are merged to create a new variational space (Space: **C**) in which a diagonalization is performed using the Davidson algorithm implemented in the FCI solver of PySCF. These processes (prediction and relearning) are repeated for the next adjacent subspaces until the total electronic energy does not vary by more than 10^{-5} Hartree.

Now that we have illustrated the functioning of the CInet algorithm, we will present the results obtained with small molecular systems across varying configuration spaces. In particular, we will compare the model's performance to the

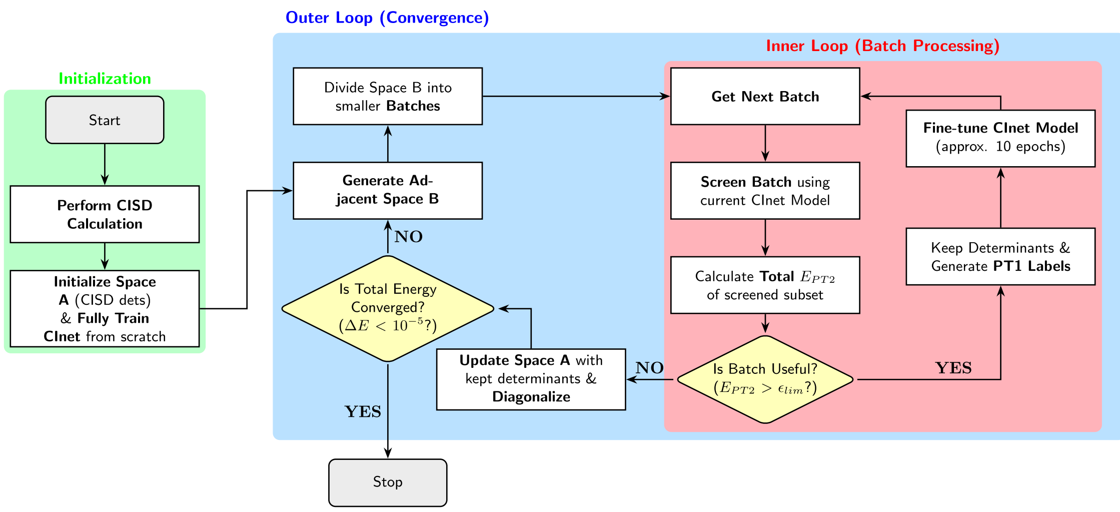

Figure 2. Flowchart of the CInet algorithm.

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standard approach CIPSI and test its robustness by examining different points along ground and excited states PEC and PES.

3. RESULTS AND DISCUSSION

To assess the inference capabilities of the method, we applied the CInet algorithm to a diverse set of chemical compounds covering a wide range of Hilbert space sizes, explored through different active spaces and three basis sets (STO-3G, 6−31G, and cc-pVDZ). The systems examined included HF, CH_4, C_2H_2, C_2H_4, C_2H_6, H_2O, NH_3, HCN and benzene, all considered at their equilibrium geometries obtained from the NIST Computational Chemistry Comparison and Benchmark Database.[43] For naphthalene, we used the optimized structure reported by Jeong *et al.*[36]

3.1. Ground State Calculations

Table 1 summarizes the absolute energies of each system at the ground state level. The CInet values are compared to two reference methods: CIPSI and FCI (or CASCI if the frozen core approximation was used). The active spaces are represented by the total number of involved electrons N and the number of accessible molecular orbitals K (Ne, Ko).

Table 1. CInet Ground State (S_0) Energies for Various Systems and Active Space Sizes (Ne, Ko)[a,b]

S_0	FCI/CASCI	CIPSI	CInet
STO-3G			
CH_4 (10e, 9o)	−39.80688	−39.80688	−39.80682
N_{det}	15,876	4809	2439
HCN (14e, 11o)	−91.84497	−91.84497	−91.84488
N_{det}	108,900	12,641	9110
C_2H_2 (14e, 12o)	−76.02559	−76.02559	−76.02558
N_{det}	627,264	22,402	24,667
C_2H_4 (16e, 14o)	−77.23665	−77.23665	−77.23663
N_{det}	9,018,009	83,409	89,087
C_2H_6 (18e, 16o)	−78.45493	−78.45492	−78.45481
N_{det}	130,873,600	883,115	684,664
6−31G			
HF (10e, 11o)	−100.11670	−100.11668	−100.11667
N_{det}	213,444	14,538	11,901
H_2O (10e, 13o)	−76.12237	−76.12236	−76.12236
N_{det}	1,656,369	44,470	25,124
NH_3 (10e, 15o)	−56.29509	−56.29507	−56.29508
N_{det}	9,018,009	208,177	257,724
CH_4 (10e, 17o)	−40.30159	−40.30156	−40.30157
N_{det}	38,291,344	1,061,227	1,079,232
HCN (10e, 18o)[†]	−93.05166	−93.05148	−93.05163
N_{det}	73,410,624	1,303,333	2,271,857
C_2H_2 (10e, 20o)[†]	−76.99858	−76.99848	−76.99854
N_{det}	240,374,016	1,680,931	2,271,857
cc-pVDZ			
H_2O (8e, 23o)[†]	−76.24036	−76.24033	−76.24032
N_{det}	78,411,025	1,428,827	1,001,587
HF (10e, 19o)	−100.22972	−100.22971	−100.22969
N_{det}	135,210,384	818,620	541,220
NH_3 (8e, 28o)[†]	−56.39846	−56.39824	−56.39835
N_{det}	419,225,625	1,796,798	2,803,234

[a]Results are compared against reference methods: FCI/CASCI and CIPSI. N_{det} denotes the number of selected determinants used to compute each excited state. All energies are presented in Hartree. [b]The symbol † refers to systems described using a frozen core approximation. For these molecules, the CASCI method was applied.

For all systems studied across various basis sets, CInet energies reach the FCI/CASCI level of theory with an error ΔE below 10^{-4} Hartree (∼0.06 kcal/mol, ∼0.003 eV). However, calculations using the minimal STO-3G basis set yield less accurate results. This reduced accuracy is attributed to the limited number of virtual molecular orbitals, which leads to a higher proportion of determinants involving electrons in occupied orbitals. We hypothesize that the binary feature space employed is insufficient to distinguish all relevant configurations in such systems.

Figure 3a−c present the variational energies of H_2O/cc-pVDZ, HF/cc-pVDZ and NH_3/cc-pVDZ as a function of the number of selected determinants. The bright green curve corresponds to the iterative CIPSI selection, while the blue points indicate the diagonalization steps performed by CInet.

Since all selective CI methods share the same asymptotic cost, which originates from the diagonalization of the Hamiltonian in the selected basis,[44] it is therefore only meaningful to compare fixed variational wave function sizes.

These figures, together with Table 1, show that CInet produces wave functions of comparable quality as CIPSI, with, overall, a similar accuracy in energy for a similar size of basis. We notice that CInet, with its simple FNN architecture, is able to identify a meaningful set of configurations contributing to the wave function, without incorporating any other explicit physical constraints than the excitation degree.

3.2. Excited State Calculations

Furthermore, because electronic excitations play a pivotal role in chemical processes, we tested the robustness of our model applying it to excited states. Table 2 reports the first neutral excitation energies ($S_0 \rightarrow T_1$) computed for each compound using the 6−31G and cc-pVDZ basis sets. As in previous benchmarks, our CInet results are compared against CIPSI[45] and FCI/CASCI reference methods for validation.

For the CIPSI calculations, we enforced the eigenvalue of the total spin-squared operator $\langle \hat{S}^2 \rangle$ to be equal to two, consistent with a triplet electronic configuration, while FCI/CASCI energies were obtained with the PySCF FCI solver.

In general, the results in Table 2 show that CInet yields excitation energies in good agreement with the FCI/CASCI references. However, the calculation for C_2H_2 did not properly converge to the T_1 eigenvalue; the obtained energy is instead closer to that of the second triplet state ($E_{T_2} = -76.74848$ Ha). In this case, the model fails to identify the most relevant configurations that would normally be selected. We suspect that the active-learning loop becomes inefficient when too few epochs are used. Aside from this outlier, the average deviation remains small, at 1.6×10^{-4} Hartree (∼0.1 kcal/mol, ∼0.004 eV).

In the following, we tested the method for different excitation energies by examining the first two valence $\pi\pi^*$ transitions—both triplet and singlet (T_1 and S_1)—in benzene and naphthalene. The active spaces composed of the π and π^* MOs consisted of 6 electrons in 12 orbitals for benzene,[9] and 10 electrons in 10 orbitals for naphthalene.[36] We additionally report the energies obtained for the first four electronic states of H_2O in the 6−31G basis set. The absolute energy of each state was determined by training a single FNN (independent of the others) using a single guess function for each target state. The results are summarized in Table 3. While the relative energies are accurate by error compensation, it is worth noting

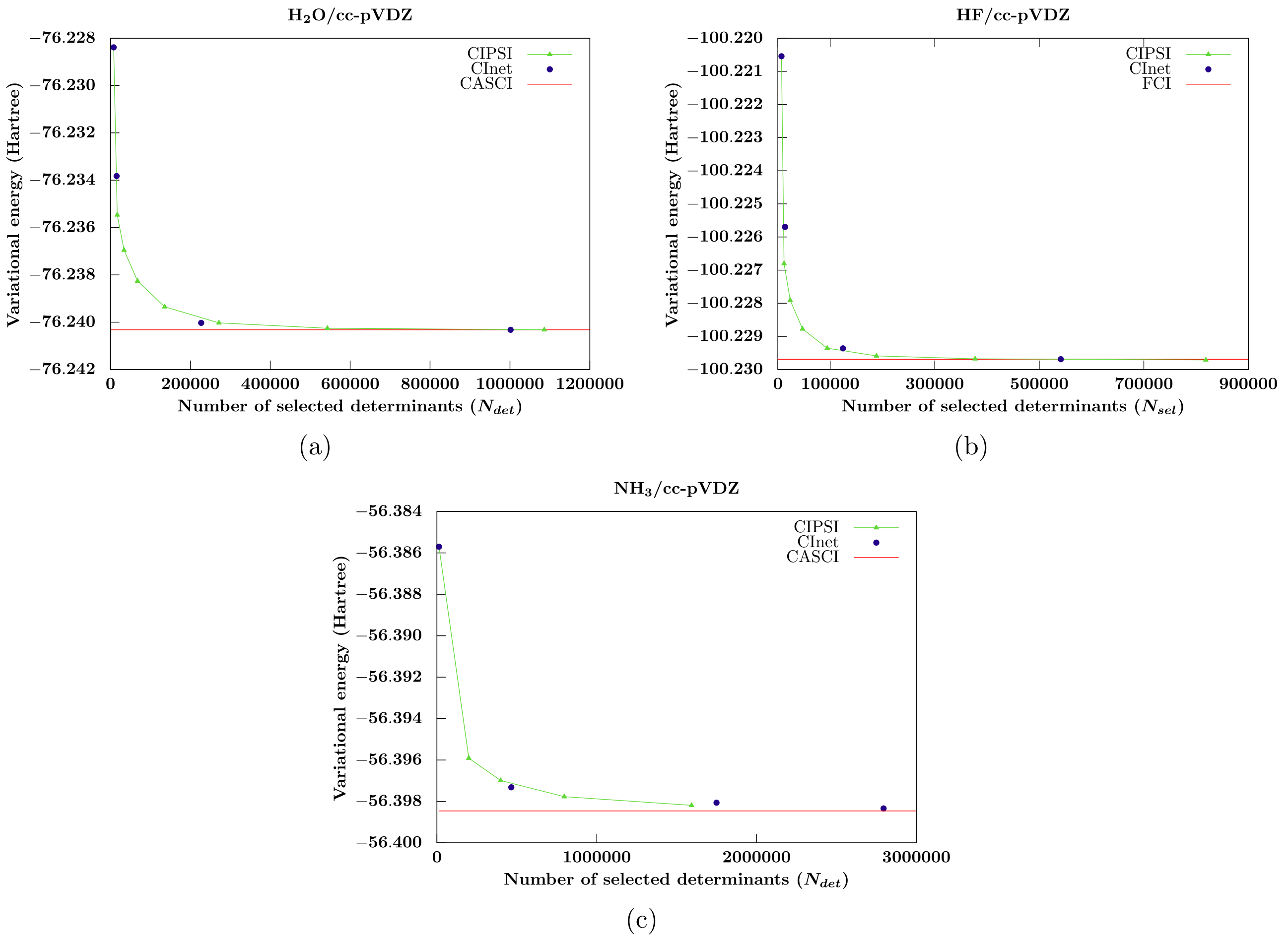

Figure 3. Variational energies (in Hartree) as a function of the number of selected determinants (N_{sel}) for the cc-pVDZ basis set. Results are shown for (a) H_2O, (b) HF, and (c) NH_3. Blue points correspond to the CInet diagonalization steps, while the bright green curves depict the CIPSI procedures. The FCI/CASCI limits are indicated by the red lines.

that the absolute energies strongly depend on the chosen value of the threshold $|c_{lim}|$, as illustrated in the SI.

This Table 3 clearly demonstrates the efficiency of the model in classifying a set of determinants for any given electronic state. The results obtained are encouraging. Once again, it is remarkable that FNNs can efficiently identify the most important determinants for any electronic state, starting only from an initial distribution of single and double excitations, without any additional physical constraints. Furthermore, we are able to substantially reduce the number of configurations required to estimate excitation energies, in line with the pioneering work of Jeong *et al.*[36]

3.3. Potential Energy Curves and Surfaces

Finally, because single-point calculations around the equilibrium geometry are insufficient to rigorously assess the robustness of the method, we further evaluated its performance on conformational changes, such as bond breaking. Figure 4 presents the PECs of HF computed with the cc-pVDZ basis set for both the ground and first excited states. The results obtained with CInet are compared against those from CIPSI and FCI calculations.

At first glance, it is evident that the CInet binding curve of ground state (S_0) is accurately reproduced, even at the dissociation limit, where strong static correlation arises due to the presence of nearly degenerate Slater determinants. The first excited state is also well described by CInet, with energies closely following the FCI reference. To quantify the accuracy of our method, we report the RMSEs in Table 4.

These observations can be extended to polyatomic molecular systems. As an illustration, Figure 5 shows the PESs of H_2O/6−31G, including both ground and excited states. Our model was tested on a grid of 500 points, where the HOH bond angle was varied from 40 to 180° in 35° increments, and the OH bond lengths were scanned between 0.5 and 2.5 Å in 10-point increments for each bond.

In Figure 5a,b, the energies obtained with CInet and CIPSI are plotted against the reference FCI values. Perfect alignment with the red curve (FCI reference) indicates accurate convergence of the modelised energies. Under this criterion, both methods exhibit comparable performance for the electronic state potential energy surfaces. This observation is consistent with the RMSE values reported in Table 4, where CIPSI and CInet achieve 2.9×10^{-2} and 3.6×10^{-2} Hartree for the ground state, and 2.7×10^{-2} and 4.9×10^{-2} Hartree for the first excited state, respectively.

4. CONCLUSIONS

In this work, we developed a binary machine learning classification method for selecting the most important Slater determinants contributing to the FCI/CASCI wave function. Similar in spirit to the CIPSI approach, our method employs a physically motivated perturbative strategy. However, it

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Table 2. CInet First Triplet State (T_1) Energies for Various Systems and Active Space Sizes (Ne, Ko)[a,b]

T_1	FCI/CASCI	CIPSI	CInet
6−31G			
HF (10e, 11o)	−99.75547	−99.75547	−99.75542
ΔE	0.36123	0.36121	0.36125
N_{det}	213,444	62,699	13,616
H_2O (10e, 13o)	−75.84617	−75.84617	−75.84610
ΔE	0.27620	0.27619	0.27625
N_{det}	1,656,369	74,603	29,898
NH_3 (10e, 15o)	−56.04532	−56.04532	−56.04530
ΔE	0.24976	0.24975	0.24978
N_{det}	9,018,009	383,528	278,214
CH_4 (10e, 17o)	−39.87143	−39.87135	−39.87136
ΔE	0.43016	0.43021	0.43021
N_{det}	38,291,344	1,134,653	1,289,120
HCN (10e, 18o)†	−92.81956	−92.81932	−92.81855
ΔE	0.23210	0.23216	0.23308
N_{det}	73,410,624	1,579,075	2,163,877
C_2H_2 (10e, 20o)†	−76.80188	−76.80172	−76.72725
ΔE	0.19671	0.19676	0.26963
N_{det}	240,374,016	1,794,092	654,263
cc-pVDZ			
H_2O (8e, 23o)†	−75.97072	−75.97065	−75.97064
ΔE	0.26964	0.26968	0.26968
N_{det}	78,411,025	1,075,794	1,154,684
HF (10e, 19o)	−99.87018	−99.87016	−99.87012
ΔE	0.35954	0.35955	0.35957
N_{det}	135,210,384	777,364	622,944
NH_3 (8e, 28o)†	−56.16069	−56.16026	−56.16025
ΔE	0.23777	0.23798	0.23810
N_{det}	419,225,625	1,920,260	3,154,876

[a]Results are compared against reference methods: CIPSI, and FCI/CASCI. The quantity ΔE denotes the neutral excitation energy between states S_0 and T_1 ($\Delta E = E_{T_1} - E_{S_0}$). N_{det} denotes the number of selected determinants used to compute each excited state. All energies are presented in Hartree [b]The symbol † refers to systems described using a frozen core approximation. For these molecules, the CASCI method was applied.

Table 3. First Triplet and Singlet Excitation Energies for Benzene (C_6H_6), Naphthalene ($C_{10}H_8$), and Water (H_2O)[a]

C_6H_6/cc-pVDZ (6e, 12o) − 48,400 det.					
excitation energy		CASCI	CInet	N_{det}	exp
	T_1	4.18	4.19	14,132	3.94[b]
	S_1	5.22	5.24	3,192	4.90[c]
$C_{10}H_8$/cc-pVDZ (10e, 10o) − 63,504 det.					
excitation energy		CASCI	CInet	N_{det}	exp
	T_1	3.28	3.44	4,239	
(ALCI[36]:4.46 eV/7,942 det)	S_1	4.46	4.63	5,212	4.66[d]
H_2O/6−31G (10e, 13o) − 1,656,369 det.					
excitation energy		FCI	CInet	N_{det}	exp
	T_1	7.52	7.52	30,270	7.20[e]
	S_1	8.26	8.26	30,902	7.41[e]
	T_2	9.52	9.52	25,188	
	T_3	9.97	9.97	31,326	

[a]N_{det} denotes the number of selected determinants used to compute each excited state. All energies are reported in eV. [b]Ref 46. [c]Ref 47. [d]Ref 48. [e]Ref 49.

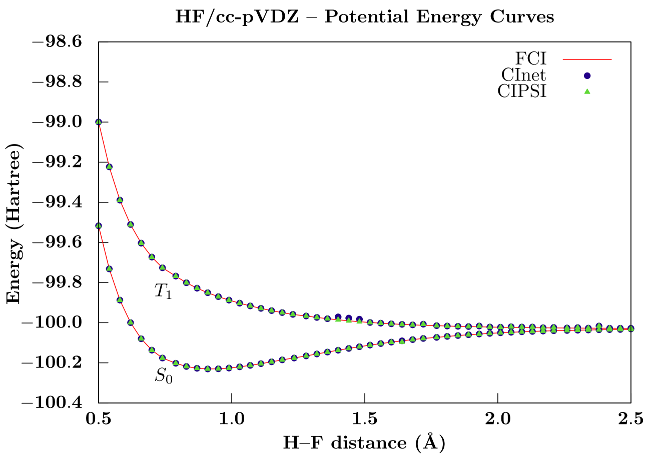

Figure 4. Potential Energy Curves (PEC) for ground and first excited states of HF/cc-pVDZ. The curves were obtained using 50 points along the H−F bond distance. Results from our CInet approach (blue points) are compared against those from CIPSI (bright green points) and FCI (red lines). All energies are given in Hartree.

distinguishes itself by leveraging Active Learning through a binary cross-entropy loss function to guide the optimal selection of determinants. The algorithm is built upon a simple FNN, which is computationally efficient and easy to train.

Across all tested configuration space sizes, the binary cross-entropy proved to be a robust metric for capturing the statistical distribution of important Slater determinants. The model accurately selected the most relevant determinants for Hilbert space sizes ranging from 2×10^4 to 5×10^8 configurations, achieving FCI/CASCI-level accuracy within 10^{-4} Hartree. This level of accuracy was also maintained for the first excited states. Notably, model performance is sensitive to the threshold value $|c_{lim}|$, which fixes the selection of determinants. This parameter is thoroughly investigated in the Supporting Information. We have also demonstrated that each neural network can be trained independently for every electronic state, starting from a guess wave function constructed from simple chemical intuition. Finally, both ground and excited state PES were accurately described at the FCI/CASCI level of theory using CInet. For instance, Figure 5c illustrates the two PES obtained for a fixed HOH angle of 145°. The hundred calculated points were interpolated using splines, yielding smooth PES. This smoothness is a first step to ensure the feasibility of computing energy gradients, a central requirement for future molecular dynamics simulations. However, the smoothness of a function alone is not sufficient; accurate gradient evaluation is essential. A detailed assessment of these forces will therefore be necessary if we intend to perform future dynamical simulations.

Looking ahead, we aim to generalize this approach to regression models. Promising preliminary work is already underway. Assuming that CInet can reliably classify the configuration space for given molecular conformations, the identified determinants could be used to construct a globally labeled data set spanning the PES. As before, Slater determinants could be encoded as binary vectors reflecting spin−orbital occupations; however, in this extended framework, the input vectors could be augmented (or just multiplied) with geometric parameters. These geometric

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Table 4. Root Mean Square Errors (RMSE) for the Two Lowest Electronic States of HF/cc-pVDZ and H_2O/6-31G[a]

RMSE (Hartree)	ground state CInet	ground state CIPSI	excited state CInet	excited state CIPSI
HF/cc-pVDZ	3.1×10^{-5}	1.9×10^{-5}	3.9×10^{-3}	2.1×10^{-3}
H_2O/6−31G	3.6×10^{-2}	2.9×10^{-2}	2.7×10^{-2}	4.9×10^{-2}

[a]The RMSEs were computed over 50 points for HF/cc-pVDZ and 500 points for H_2O/6-31G. All values are given in Hartree.

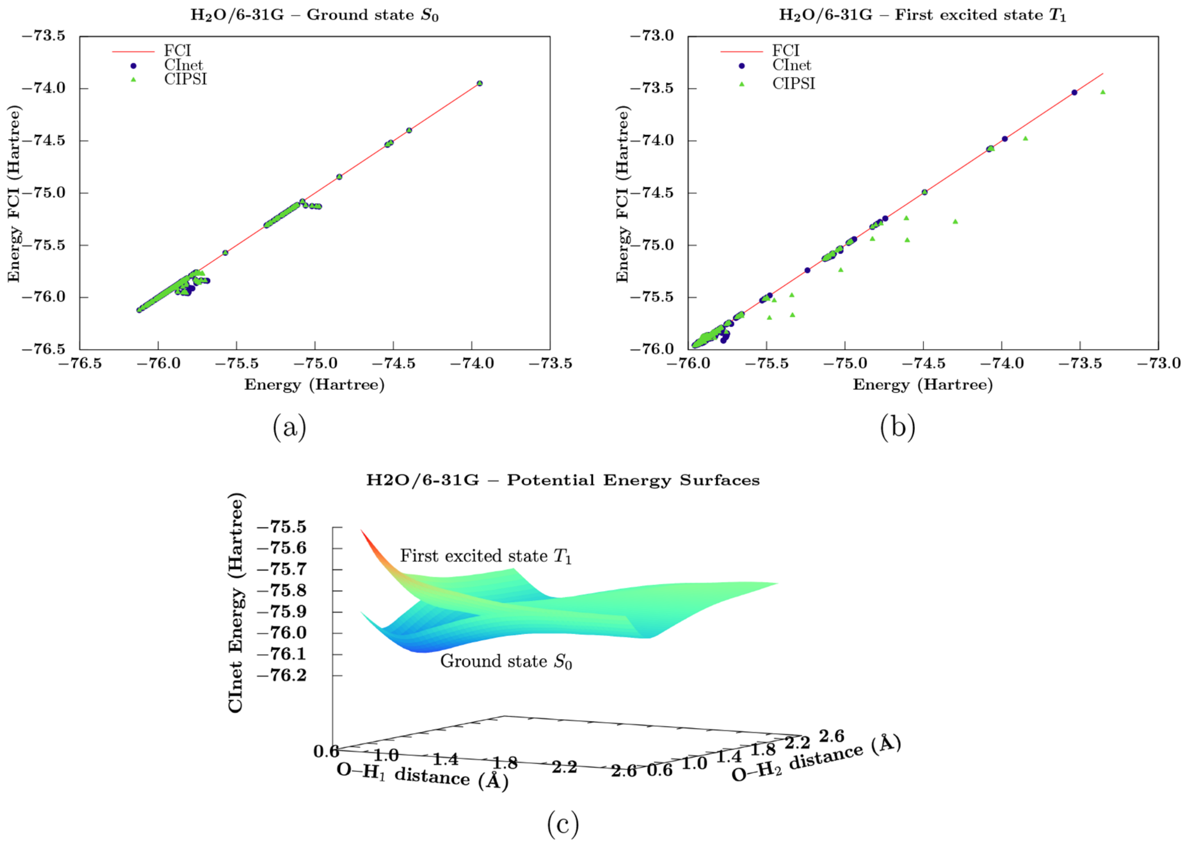

Figure 5. FCI energies against energies computed through CIPSI and CInet methods: (a) H_2O/6−31G Ground state, (b) H_2O/6−31G First excited state. (c) Potential Energy Surfaces (PES) for the two H_2O/6−31G lowest electronic states computed with our CInet method at a fixed HOH bond angle of 145°. These PES slices were generated from 100 calculated points, which were interpolated using 30 splines along each coordinate direction. All energies are given in Hartree.

features are crucial for enabling the model to track energy (or wave function) evolution with respect to molecular structure.

This methodology opens the door to studying small-molecule dynamics at the FCI/CASCI level in future work. A particularly promising application would be geometry optimization. Since the geometric information—*e.g.*, interatomic distances or Root Mean Square Deviation (RMSD)—is already embedded in the input, the FNN can be adapted for optimization tasks with minimal modification. Another potential application lies in the binary classification of product eigenfunctions, which could facilitate vibrational mode selection—a central aspect of nuclear quantum dynamics.

Nonetheless, there is still room for improvement in determinant selection. Some of the current limitations may be mitigated by incorporating belief functions[50] or probabilistic "*filters*" into the acceptance criteria for determinant classification. Another promising direction lies in devising more efficient sampling strategies for exploring the configuration space.[51]

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The full CInet code is openly accessible on GitHub at: https://github.com/b4st13nc/cinet.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.5c01652.

Optimal hyperparameter search; inference performance; influence of the $c_{\lim}$ threshold; and additional details on all CInet calculations performed (PDF)

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Author Contributions

B.C. conceived the original idea together with B.H. He developed both the Python and C codes, carried out the excited-state calculations, and contributed to the full writing of this manuscript. M.E.H. performed the hyperparameter screening and the ground-state calculations. B.H. and M.E.H. also critically reviewed the manuscript.

Notes

The authors declare no competing financial interest.

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